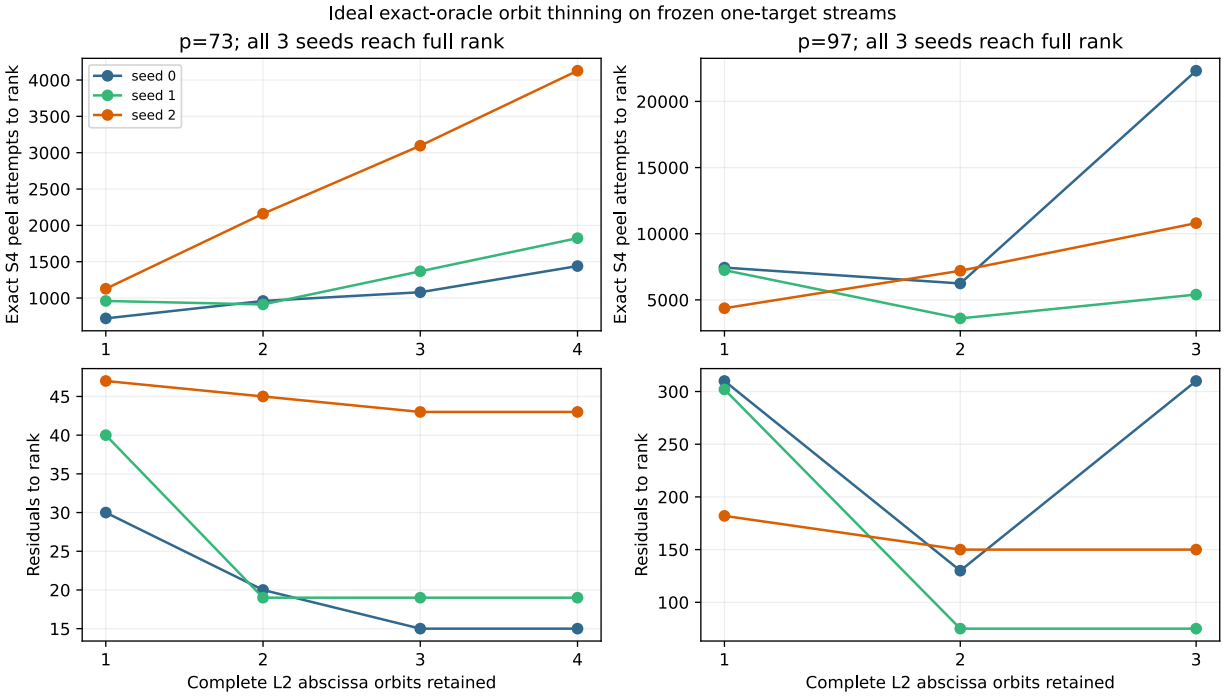
Orbit-complete line-2 thinning in quartic-twist index calculus

23 October 2026. Preregistered ideal exact-oracle survey on all 21 orbit-complete line-2 subsets of the six public one-target p=73/p=97 streams. The full source-linked report, native input records and byte-identical A/A run are in `K1_ORBIT_THINNING.md` and the two JSON records beside it. This is a group-oracle rank and peel-count result, not a polynomial speed comparison.

Construction and independent checks

Every line-2 abscissa orbit has 12 physical factors and one folded log column. The k=1 collector asks for up to 24 signed S4 peels per orbit per residual. Keep all line-0 factors, then retain the first j complete line-2 orbits by smallest physical index, for every j=1..4 at p=73 and j=1..3 at p=97. The complete scalar-keyed signed set on every query matches the earlier independently audited point-keyed census digest. Filtering that set by line-2 orbit gives the exact subset relations. Folded rows are added in sorted order until full rank. An independent Python Fp4 elliptic-curve implementation checks every solved factor representative and target by scalar multiplication. All 21 cases reach rank; the A/A JSON is byte-identical.



Frozen grid

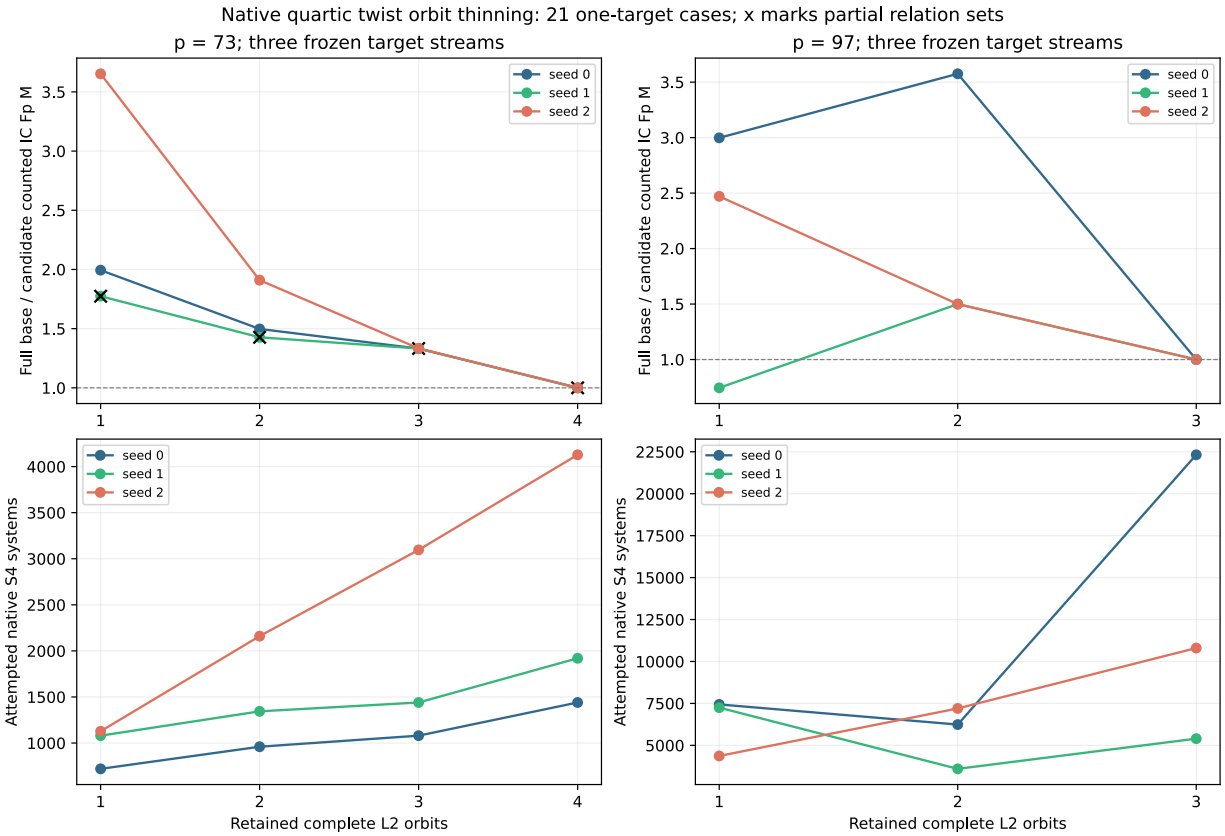
p/seed	j	L2 size	cols	queries	prefix hits	S4 peels
73/0	1	12	5	30	6	720
73/0	2	24	6	20	6	960
73/0	3	36	7	15	7	1,080
73/0	4	48	8	15	8	1,440
73/1	1	12	5	40	6	960
73/1	2	24	6	19	8	912
73/1	3	36	7	19	9	1,368
73/1	4	48	8	19	10	1,824
73/2	1	12	5	47	5	1,128
73/2	2	24	6	45	6	2,160
73/2	3	36	7	43	11	3,096
73/2	4	48	8	43	15	4,128
97/0	1	12	4	310	4	7,440
97/0	2	24	5	130	8	6,240
97/0	3	36	6	310	16	22,320
97/1	1	12	4	302	4	7,248
97/1	2	24	5	75	5	3,600
97/1	3	36	6	75	6	5,400
97/2	1	12	4	182	4	4,368
97/2	2	24	5	150	5	7,200
97/2	3	36	6	150	6	10,800

Interpretation

The lowest ideal S4 peel count in each seed is 1.5 to 3.66 times below the full-base count, but the best orbit count varies by seed. The p=73 seed-one full-base exact oracle stops at query 19, one query before the partial polynomial collector: it uses two relations missed by a failed native S4 system. These are exact requested-system counts, not solver field work or runtime. No native polynomial run has yet tested the thinned bases. A prospective all-j native grid with independent replay, full accounting and matched rho is the next measurement; selecting a best j after this survey cannot justify a speed claim.

Later native validation (23 October)

The complete 21-case native polynomial grid has since been run on these same frozen streams. Every candidate reaches full anchored rank and recovers 271 under independent point-keyed/BSGS replay; first/A/A and six full-base source-parity checks pass. Seventeen processed relation sets are exact. All four p=73 seed-one subsets retain two missing relations from the same failed S4 system. The source-linked `K1_THIN_NATIVE_REPORT.md` and its PDF contain the counted-field comparison and every negative or partial outcome. The exact-oracle survey above remains its original hypothesis screen, not candidate polynomial work.



Native field-work and attempted-system paths vary by seed and orbit count; p=97 seed-one with one orbit is a recorded regression. The native report supplies every per-case counted cost and rho gap.